

Mathematical Foundations of Image Synthesis

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1 Introduction

Image synthesis is the process of generating images from mathematical and algorithmic descriptions. Modern graphics systems rely on **precise mathematical models** and efficient computation. *Even small numerical errors* can result in visible artifacts in rendered images.

Typical formulations involve scalar values such as x , indexed parameters like I_{ij} , and exponential terms e^{x^2} . These elements are combined to simulate realistic lighting behavior.

2 Core Components

2.1 Rendering Pipeline

2.2 Execution Order

3 Mathematical Modeling

4 Performance Comparison

5 Visual Illustration

6 Conclusion